

12.2 Solving Problems With Lengths of Two Secants

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems that involve the lengths formed by the intersection of two secants.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the relationship of the lengths formed by the intersection of two secants? - How can you find the lengths of segments formed by the intersection of two secants?			
 Lesson Narrative	This lesson will introduce the student to a secant line and its properties. Students will start with an exploration of given segment lengths and will complete a table about the lengths formed by the intersection of two secants. Students will work with a partner to make a conjecture and then write a formula about the lengths formed by the intersection of two secants. Students will be guided through a series of problems, including real-world examples. Students will conclude by trying problems on their own.			
 Component Summary	12.2.1 Warm-Up (~5 min.)	Would You Rather? with gold cubes (<i>SE p. 238</i>)	12.2.3 Guided Practice (10-15 min.)	Solving problems involving intersecting secant lengths (<i>SE p. 239</i>)
	12.2.2 Exploration (10-15 min.)	Explore intersecting secant lengths theorem (<i>SE p. 238</i>)	12.2.4 Your Turn! (5-10 min.)	Solving problems involving intersecting secant lengths independently (<i>SE p. 241</i>)
	12.2.5 Wrap-Up (~5 min.)	Error Analysis of student work finding lengths of intersecting secants (<i>SE p. 241</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Secant <u>Additional Terms:</u> N/A		(Optional) Chart Paper (Optional) Markers or Colored Pencils	N/A

 Teacher Tips	Common Misconceptions • Students may confuse a secant line and a tangent line. • Students may confuse multiplying and adding in the formula.	Things to Consider • Consider illustrating the theorem on a piece of chart paper that will be left hanging in the classroom as a reference for students. • If the students use an interactive notebook, consider having questions 1a-c from 12.2.2 Exploration available for them to paste into their notebook or have them copy questions 1a-c in their notebooks and color code the appropriate parts in the diagrams.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 12.2.2 Exploration provides a structured and interactive opportunity for students to revise and refine both their ideas, and their verbal and written output. This routine provides a purpose for student conversation, as well as fortifies output. Throughout this process, students should be pressed for details, and encouraged to press each other for details.	MA.K12.MTR.4.1: Discussion and Reflection 12.2.2 Exploration allows the students to communicate mathematical ideas, vocabulary, and methods effectively. 12.2.2 Exploration will develop students' ability to justify methods and compare their responses to the responses of their peers.
 Differentiate Instruction	GeoGebra and other online tools provide digital manipulatives for engaging in content like chords, secants, and circles. 12.2.2 Exploration or 12.2.3 Guided Practice might be opportunities for students to engage with a digital manipulative to make connections between the lengths of secants and their relationship to each other.	
Lesson Notes:		

12.2.1 Warm-Up: Would You Rather?

Component Overview: Students are asked to determine which cube of gold they would rather have.



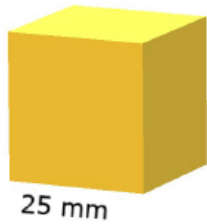
12.2.1 Warm-Up: Would You Rather?

1. Would you rather have ...

A cube of gold, 25 millimeters(mm) on each side ...

OR

Two cubes of gold, one is 24 mm per side, one is 1 mm per side ...



$$25^3 = 15,625 \text{ mm}^3$$



$$24^3 + 1^3 = 13,825 \text{ mm}^3$$



Provide the currency exchange rates for gold, silver, the US Dollar, cryptocurrency, and other country's currency for an engaging discussion on saving or investing money.

2. Explain your choice.

Answers will vary. I would choose the 25mm cube of gold because it weighs more.

12.2.2 Exploration: Intersecting Secant Lengths

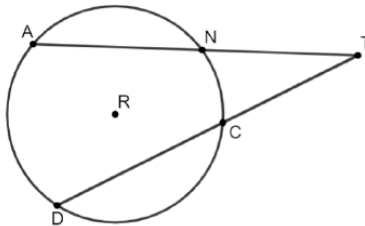
Component Overview: Students begin working with intersecting secant lengths. The outcome of this component is for students to be able to discover the formula used to find the lengths of intersecting secant lines. Consider giving students opportunities to analyze errors, engage in math talk, and collaborate.



12.2.2 Exploration: Intersecting Secant Lengths

A **secant** is a line that passes through two points on a circle. In the interior of the circle, a secant line forms a chord.

- Circle R is shown with points A , N , D , and C on the circle. Secants $\overline{AT}$ and $\overline{DT}$ intersect at point T outside the circle. $AN = 6$, $CD = 9$, $NT = 8$, and $CT = 7$.



- Complete the table.

Product	Work	Answer
$AN \cdot NT$	$AN \cdot NT = (6)(8)$	48
$AN \cdot AT$	$AN \cdot AT = (6)(14)$	84
$AT \cdot NT$	$AT \cdot NT = (14)(8)$	112
$DC \cdot CT$	$DC \cdot CT = (9)(7)$	63
$DC \cdot DT$	$DC \cdot DT = (9)(16)$	144
$DT \cdot CT$	$DT \cdot CT = (16)(7)$	112

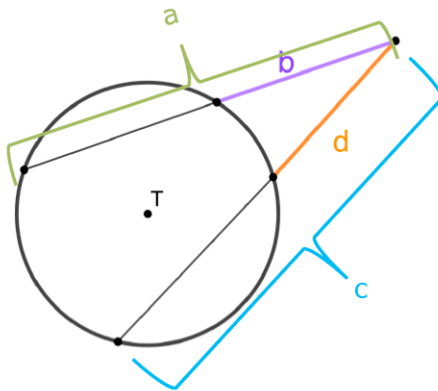


MA.K12.MTR.5.1: Patterns and Structure

Encourage students to use patterns and structure to help understand and build connection between the intersecting secant lines. This will support your students to develop generalizations based on similarities found among questions. This will also help students recognize patterns in the world around them and connect these patterns to mathematical concepts.

12.2.2 Exploration: Intersecting Secant Lengths (continued)

- b. Discuss the following with your partner.
- What do you notice about your answers?
 - What do you notice about the secant segments for the rows that have the same answer?
- c. With your partner, use the diagram of Circle T to write a formula that can be used to find the segment lengths of intersecting secants.



$$ab = cd$$



If two secants intersect outside a circle, then the product of the lengths of one secant segment and its external secant segment equals the product of the lengths of the other secant segment and its external secant segment.



Provide sentence frames to facilitate strong mathematical discussion:

- “When two secants intersect...”
- “The product of the lengths of one secant...”
- “When two segments of a secant are multiplied...”

After discussion, prompt students to write a summary of their findings.



Consider challenging students to write a formula if a and c were not the whole length of the secant lines, but the part of the secant line inside the circle. $[b(a + b) = d(c + d)]$

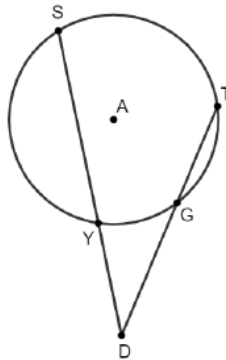
12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths

Component Overview: Students are guided through using the formula generated in 12.2.2 Exploration to solve for variables and missing lengths. Students are expected to apply this knowledge to real-world situations.



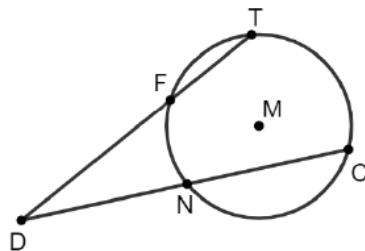
12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths

- Circle A with secants $\overline{SD}$ and $\overline{TD}$ is shown. $\overline{SD}$ and $\overline{TD}$ intersect at point D outside the circle so that $SY = 5$, $DY = 3$, $TG = x$, and $DG = 4$. Solve for x .



$$3(3 + 5) = 4(4 + x) \quad x = 2$$

- Circle M is shown, where points T , F , N , and C are on circle M . Secants $\overline{TD}$ and $\overline{CD}$ intersect at point D outside the circle such that $TF = 7$, $FD = 9$, $CN = 5x$, and $ND = 4x$. Determine the value of x .



$$4x(9x) = 9(16) \quad x = 2$$



Are you given the lengths to the formula as written? $\overline{DY}(\overline{DY} + \overline{SY}) = \overline{DG}(\overline{DG} + \overline{TG})$. If not, how can you manipulate the lengths you are given to work in this formula?



Some students may need assistance when solving for the value of x . Consider modeling an example for students by sharing your thought process out loud as you solve questions 1 and 2.



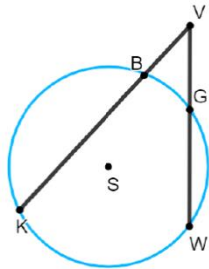
Consider showing students how to color code the information given on the picture using colored pencils or highlighters. This will help students to identify the segment lengths given.

12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths (continued)

3. Leon Sinks Geologic Area is located in Apalachicola National Forest. The area has 18 sinkholes, many of which create a cave system. A park ranger mapped a circular sinkhole. His diagram



is shown, where circle S has secants $\overline{VK}$ and $\overline{VW}$ such that points B and G are on circle S and point V is outside the circle. $KB = (4x + 4.25)$ meters (m), $VB = 12.75$ m, $GW = 3x + 5.5$ m, and $GV = 14$ m. Find the distance, in meters, from point K to point B .



$$12.75(4x + 4.25 + 12.75) = 14(3x + 5.5 + 14) \quad x = 6.25$$

$$KB = 4(6.25) + 4.25 = 29.25m$$



Allow students to engage in meaningful dialogue related to this real-life situation and challenge students to question the accuracy of this model.

12.2.4 Your Turn!

Component Overview: Students are given the opportunity to take what was practiced in 12.2.3 Guided Practice and apply independently. Probe students as they work using assessing and critical thinking questions to check for understanding, to challenge students, and to help students who are struggling. Consider a peer review process for students to check each other's work as an added layer of support.

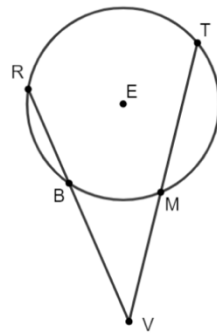


12.2.4 Your Turn!

- Circle E has points R , F , G , and M on the circle. Secants $\overline{RZ}$ and $\overline{MZ}$ intersect at point Z outside the circle. Point G is on $\overline{MZ}$, and point F is on $\overline{RZ}$. $MG = 10$, $GZ = 8$, $RF = (2x - 17)$, and $FZ = 9$. Solve for x .
 $9(2x - 17 + 9) = 8(18)$ $x = 12$
- Tammie was calculating the distance between her favorite places to visit in Epcot. Circle E represents the World Showcase where R , T , M , and B are on the circle and point V is outside the circle at Spaceship Earth. Tammie knows that $RB = 975$ feet (ft), $BV = 1425$ ft, and $VM = 1250$ ft. What is the length of $\overline{MT}$, in feet?

$$1425(1425 + 975) = 1250(1250 + MT)$$

$$MT = 1486 \text{ ft}$$



Encourage students to use a diagram or modeling for making sense of question 1. Provide chart paper for a small group who may be able to engage with question 1 as a group, constructing an Anchor Chart while working.

12.2.5 Wrap-Up: Error Analysis

Component Overview: Students complete an Error Analysis by finding the mistake made and writing a note to the students explaining the mistake made when solving for x .

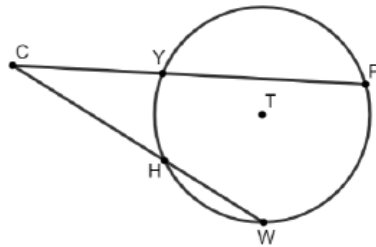


12.2.5 Wrap-Up: Error Analysis

1. Donald was finding the value of x in circle T , where Y , P , W , and H are points on the circle. He knows that $\overline{CP}$ and $\overline{CW}$ are secants of the circle, where $CY = 4$, $YP = x - 3$, $CH = 5$, and $HW = x - 6$. His work is shown.

Donald's Work

$$\begin{aligned} 4(x - 3) &= 5(x - 6) \\ 4x - 12 &= 5x - 30 \\ x &= 18 \end{aligned}$$



Write a note to Donald explaining the mistake he made when solving for x .

Answers will vary.

Donald,

You need to multiply CY by the entire segment length, CP , and multiply CH by the entire segment length, CW , as shown below:

$$\begin{aligned} 4(x - 3 + 4) &= 5(x - 6 + 5) \\ x &= 9 \end{aligned}$$



Clarify, Critique, Correct

The intent is to prompt student reflection with an incorrect, incomplete, or ambiguous written argument or explanation, and for students to improve upon the written work by correcting errors and clarifying meaning. This routine fortifies output and engages students in meta-awareness. Consider demonstrating with a think-aloud and press for details when necessary.